

Basic Study of Copula Theory and its application in Copula Opinion Pooling model

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What are **Copulas**?

They are functions that join or 'couple' multivariate distribution functions to their one-dimensional marginal distribution functions.

Alternatively, copulas are multivariate distribution functions whose one-dimensional margins are uniform on the interval $(0,1)$.
(This would make more sense a bit later)

H-volume & grounded 2-increasing functions

Definition 1:

Let S_1 and S_2 be nonempty subsets of $\overline{\mathbb{R}}$, and let H be a two-place real function such that $\text{Dom}H = S_1 \times S_2$. Let $B = [x_1, x_2] \times [y_1, y_2]$ be a rectangle all of whose vertices are in $\text{Dom}H$. Then the *H-volume* of B is given by

$$V_H(B) = H(x_2, y_2) - H(x_2, y_1) - H(x_1, y_2) + H(x_1, y_1).$$

Definition 2:

A 2-place real function H is *2-increasing* if $V_H(B) \geq 0$ for all rectangles B whose vertices lie in $\text{Dom}H$.

H-volume & grounded 2-increasing functions

From the above two definitions we have,

Lemma 1:

Let S_1 and S_2 be nonempty subsets of $\overline{\mathbb{R}}$, and let H be a 2-increasing function with domain $S_1 \times S_2$. Let x_1, x_2 in S_1 with $x_1 \leq x_2$, and let y_1, y_2 be in S_2 with $y_1 \leq y_2$. Then the functions

$$t \mapsto H(t, y_2) - H(t, y_1)$$

$$t \mapsto H(x_2, t) - H(x_1, t)$$

are non-decreasing on S_1 & S_2 respectively.

H-volume & grounded 2-increasing functions

Definition 3:

Let S_1 has a least element a_1 and S_2 has a least element a_2 . We say that a function $H : S_1 \times S_2 \rightarrow \mathbb{R}$ is *grounded* if

$$H(a_1, y) = 0 \quad \forall y \in S_2 \quad \& \quad H(x, a_2) = 0 \quad \forall x \in S_1.$$

Lemma 2:

Let S_1 and S_2 be nonempty subsets of $\overline{\mathbb{R}}$, and let H be a *grounded 2-increasing* function with domain $S_1 \times S_2$. Then H is non-decreasing in each argument.

H-volume & grounded 2-increasing functions

Definition 4:

Let S_1 has the greatest element b_1 & S_2 has the greatest element b_2 .
We then say that a function H has margins, F and G given by

$$F(x) = H(x, b_2) \quad \forall x \in S_1 \text{ where, } \text{Dom}F = S_1$$

$$G(y) = H(b_1, y) \quad \forall y \in S_2 \text{ where, } \text{Dom}G = S_2.$$

Now we can conclude that,

Lemma 3:

Let S_1 and S_2 be nonempty subsets of $\overline{\mathbb{R}}$, and let H be a *grounded 2-increasing* function, with margins, whose domain is $S_1 \times S_2$. Let (x_1, y_1) and (x_2, y_2) be any points in $S_1 \times S_2$. Then

$$|H(x_2, y_2) - H(x_1, y_1)| \leq |F(x_2) - F(x_1)| + |G(y_2) - G(y_1)|.$$

Definition 5:

A two-dimensional subcopula (or 2-subcopula) is function C' with the following properties:

1. $\text{Dom}C' = S_1 \times S_2$ where S_1 and S_2 are subsets of $\mathbf{I}=[0,1]$ such that both S_1 and S_2 contains $\{0, 1\}$.
2. C' is a *grounded 2-increasing* function.
3. For every $u \in S_1$ & $v \in S_2$, we have $C'(u, 1) = u$ and $C'(1, v) = v$.

Definition 6:

A two-dimensional copula (or 2-copula) is a 2-subcopula whose domain is $\mathbf{I}^2$.

Note: $V_C([0, u] \times [0, v]) = C(u, v)$.

Fréchet Hoeffding bounds

Theorem 1:

Let C' be a subcopula. Then for every (u, v) in $\text{Dom}C'$,

$$\max(u + v - 1, 0) \leq C'(u, v) \leq \min(u, v)$$

where these bounds are called **Fréchet Hoeffding bounds**.

Theorem 2:

Let C' be a subcopula. Then for every (u_1, v_1) & $(u_2, v_2) \in \text{Dom}C'$,

$$|C'(u_2, v_2) - C'(u_1, v_1)| \leq |u_2 - u_1| + |v_2 - v_1|$$

i.e., C' is uniformly continuous on its domain.

Definition 7:

Let C be a copula, and let a be any number in $\mathbf{I}$.

1. The *horizontal section* of C is the function from $\mathbf{I}$ to $\mathbf{I}$ given by $t \mapsto C(t, a)$.
2. The *vertical section* of C is the function from $\mathbf{I}$ to $\mathbf{I}$ given by $t \mapsto C(a, t)$.
3. The *diagonal section* of C is the function δ_C from $\mathbf{I}$ to $\mathbf{I}$ defined by $\delta_C = C(t, t)$.

Corollary 1:

The *horizontal*, *vertical*, and *diagonal* sections of copula C are all non-decreasing and uniformly continuous on $\mathbf{I}$.

Contour Diagrams

A simple but useful way to present the graph of copula is with a contour diagram i.e., the graphs of its level sets $\subseteq \mathbf{I}^2$.

Let $t \in \mathbf{I}$, then

$$L_t(C) = \{(u, v) \in \mathbf{I}^2 | C(u, v) = t\}.$$

Note:

Since $C(1, t) = C(t, 1) = t \Rightarrow$ For any $t \in \mathbf{I}$, $(1, t)$ & $(t, 1) \in L_C(t)$.

Sklar's Theorem

Definition 8:

A *distribution function* is a function F with domain $\overline{\mathbb{R}}$ such that

1. F is non-decreasing,
2. $F(-\infty) = 0$ and $F(\infty) = 1$.

Definition 9:

A *joint distribution function* is a function H with domain $\overline{\mathbb{R}}^2$ such that

1. H is 2-increasing,
2. $H(x, -\infty) = H(-\infty, y) = 0$, and $H(\infty, \infty) = 1 \forall x, y \in \overline{\mathbb{R}}$.

Note:

$F(x) := H(x, \infty)$ & $G(y) := H(\infty, y)$ are margins of H .

Theorem 3: Sklar's Theorem

Let H be a joint distribution function with margins F & G . Then $\exists$ a copula C such that for all $\forall x, y \in \overline{\mathbb{R}}$,

$$H(x, y) = C(F(x), G(y)).$$

If F and G are continuous, then C is unique; otherwise, C is uniquely determined on $\text{Ran}F \times \text{Ran}G$. Conversely, if C is a copula and F and G are distribution functions, then the function H defined above is a joint distribution function with margins F and G .

We need two lemmas to prove this theorem.

Lemma 4:

Let H be a joint distribution function with margins F and G . Then there exists a unique subcopula C' such that

1. $\text{Dom}C' = \text{Ran}F \times \text{Ran}G$,
2. For all x, y in $\overline{\mathbb{R}}$, $H(x, y) = C'(F(x), G(y))$.

Lemma 5:

Let C' be a subcopula. Then $\exists$ a copula C such that $C(u, v) = C'(u, v)$ for all (u, v) in $\text{Dom}C'$; i.e., any subcopula can be extended to a copula. The extension is generally non-unique.

Sklar's Theorem

Definition 10:

Let F be a distribution function. Then a *quasi-inverse* of F is any function $F^{(-1)}$ with domain $\mathbf{I}$ such that

1. If $t \in \text{Ran}F$, then $F^{(-1)}$ is any number x in $\overline{\mathbb{R}}$ such that $F(x) = t$, i.e., for all t in $\text{Ran}F$,

$$F(F^{(-1)}) = t;$$

2. If $t \notin \text{Ran}F$, then

$$F^{(-1)} = \inf\{x | F(x) \geq t\} = \sup\{x | F(x) \leq t\}$$

Corollary 2:

Let H, F, G and C' be as in Lemma 4, and let $F^{(-1)}$ and $G^{(-1)}$ be quasi-inverses of F and G , respectively. Then for any (u, v) in $\text{Dom}C'$,

$$C'(u, v) = H(F^{(-1)}(u), G^{(-1)}(v)).$$

Any Copula can be thought of as restriction to $\mathbf{I}^2$ of a joint distribution function whose margins are uniformly distributed functions on $[0,1]$.

Copula Opinion Pooling

1. This model is an extension of Black-Litterman to a generic non-normal prior market distribution and non-normal views.
2. The Black-Litterman model allows the portfolio manager to smoothly blend his subjective views on the market with a prior market distribution.
3. The major drawback of BL methodology is that it works only when the prior market distributions and views distributions are assumed to be normal.

Copula Opinion Pooling

Let $\mathbf{M}$ be a set of random variables that determines the randomness in the market.

$$\mathbf{M} \sim (f_{\mathbf{M}}, F_{\mathbf{M}}, \phi_{\mathbf{M}}).$$

Then we have views as a set of $K \leq N$ statements on generic linear combinations of the market, and these linear combinations are represented by a $K \times N$ dimensional 'pick' matrix P .

Therefore the views are the following K -variate random vector:

$$\mathbf{V} \equiv \mathbf{P}\mathbf{M}.$$

The manager expresses the generic k -th view in terms of a subjective univariate distribution, which is represented by its pdf, cdf, or cf:

$$f_{\hat{V}_k}, F_{\hat{V}_k}, \phi_{\hat{V}_k}.$$

On the other hand, market prior distribution $\mathbf{M}$ induces a structure on the distributions of views. Thus we have the marginal distribution of the generic k -th view reads as follows in terms of its cf:

$$V_k \sim \phi_{V_k}(\omega) = \phi_{\mathbf{M}}(\mathbf{p}_k' \omega),$$

where $\mathbf{p}_k$ denotes the k -th row of the pick matrix $\mathbf{P}$.

Copula Opinion Pooling

we first define the posterior pdf of the generic k -th view as a weighted average of the market-induced prior pdf (f_{V_k}) and the subjective pdf ($f_{\hat{V}_k}$) that expresses the view:

$$\tilde{V}_k \sim f_{\tilde{V}_k} \equiv (1 - c_k)f_{V_k} + c_k f_{\hat{V}_k}$$

We now first compute the copula of the views induced by the market prior, and then trace back the joint posterior distribution for the views.

In this model, Copula played an important role in obtaining the joint distribution function from its given marginals.

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Thank You